ACS APPLIED MATERIALS & INTERFACES

www.acsami.org

Research Article

Facile Synthesis of Defect-Modified Thin-Layered and Porous g-C_3N_4 with Synergetic Improvement for Photocatalytic H_2 Production

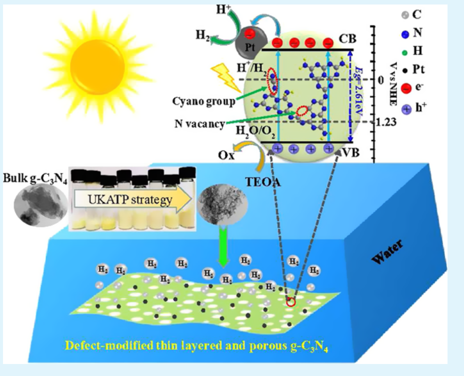
Yanbin Huang, Jun Liu, Chao Zhao, Xiaohao Jia, Mengmeng Ma, Yuanyuan Qian, Cheng Yang, Kong Liu, Furui Tan, Zhijie Wang,* Xiaobao Li,* Shengchun Qu,* and Zhanguo Wang

Cite This: https://dx.doi.org/10.1021/acsami.0c14262

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Modulating and optimizing the diverse parameters of photocatalysts synergistically as well as exerting these advantages fully in photocatalytic reactions are crucial for the sufficient utilization of solar energy but still face various challenges. Herein, a novel and facile urea- and KOH-assisted thermal polymerization (UKATP) strategy is first developed for the preparation of defect-modified thin-layered and porous g-C_3N_4 (DTLP-CN), wherein the thickness of g-C_3N_4 was dramatically decreased, and cyano groups, nitrogen vacancies, and mesopores were simultaneously introduced into g-C_3N_4. Importantly, the roles of thickness, pores, and defects can be targetedly modulated and optimized by changing the mass ratio of urea, KOH, and melamine. This can remarkably increase the specific area, improve the light-harvesting capability, and enhance separation efficiency of photoexcited charge carriers, strengthening the mass transfer in g-C_3N_4. Consequently, the photocatalytic hydrogen evolution efficiency of the DTLP-CN (1.557 mmol h^{-1} g^{-1}, $\lambda > 420$ nm) was significantly improved more than 48.5 times with the highest average apparent quantum yield (AQY) of 18.5% and reached as high as 0.82% at 500 nm. This work provides an effective strategy for synergistically regulating the properties of g-C_3N_4, and opens a new horizon to design g-C_3N_4-based catalysts for highly efficient solar-energy conversion.



KEYWORDS: *water splitting, facile strategy, defect, thickness, synergistically regulate*

1. INTRODUCTION

As inspired by natural photosynthesis, development of an efficient solar water-splitting system has boosted intensive research interest, which is one of the most promising technology to address the global energy and environmental crisis.[1,2] Since the photoelectrochemical (PEC) water splitting was first discovered in 1972, great progress has been achieved in seeking and optimizing efficient photocatalysts.[3−9] Among various visible-light-responding photocatalysts, metal-free g-C_3N_4 has attracted extensive attention, owing to its abundance on earth, nontoxicity, high chemical and thermal stability, and suitable band structure for both hydrogen and oxygen evolution reactions.[10−13] Such merits also endow it with wide applications in other photocatalytic reactions, such as CO_2 reduction, organic photosynthesis, and pollution degradation.[14−17] However, fast photoexcited charge recombination, low specific surface area, and insufficient visible-light-harvesting capacity of pristine g-C_3N_4 seriously impede its practical application.

To address the aforementioned challenges, various strategies, such as morphological and structural engineering,[18−22] element doping,[23−27] exfoliation, and coupling with other materials,[28−36] have been proposed to improve the performance of g-C_3N_4. Especially, the introduction of N defects into the framework of g-C_3N_4 has been proposed as an effective way to enhance its photocatalytic activity.[19,20,25] It was demonstrated that the introduced defects not only acted as active sites for photocatalytic reactions but also functioned to regulate the band structure by generating the defect energy level in the band gap. Besides, the defects can act as trapping sites for photoexcited charge carriers and remarkably suppressed the recombination of electron−hole pairs. Moreover, it was validated that the thickness and the porous structure were also closely related to the photocatalytic activity of g-C_3N_4. Decreasing the thickness of g-C_3N_4 can greatly increase its specific area and reactive sites and shorten the diffusion path of photoexcited charge carriers.[30,37,38] Meanwhile, construction of a porous structure into g-C_3N_4 can significantly enhance the mass transfer and improve its light-absorption capacity by the multiple scattering effects.[39−41] As a result, the photocatalytic performance of g-C_3N_4 was dramatically enhanced. It suggests that by modifying the thickness and introducing pores and defects into g-C_3N_4, the encouraging improvement in photo-

Received: August 10, 2020
Accepted: October 30, 2020

catalytic properties could be expected. However, to the best of our knowledge, most of the reports were almost devoted to single or two roles of thickness, pores, and defects. A comprehensive study is still lacking on the synergetic effects of the above three roles on improving the photocatalytic activity of g-C_3N_4.[19,26,27,35] As the optimal efficiency of the solar water-splitting reaction requires a distinct balance among diverse parameters, such as light-absorption capacity, surface active sites, separation, and transformation of photoexcited carriers, the search for a facile strategy for delicately introducing and modulating the three roles is highly desirable.

In this work, we first develop a facile urea (U) and KOH-assisted thermal polymerization (UKATP) strategy to fabricate defect-modified thin-layered and porous g-C_3N_4 (DTLP-CN), wherein the thickness of g-C_3N_4 is dramatically decreased, and cyano groups, nitrogen vacancies, and mesoporous features are simultaneously introduced into g-C_3N_4. Especially, the roles of thickness, pores, and defects can be targetedly modulated and optimized by changing the mass ratio of U, KOH, and melamine (M). The experimental results and density functional theory (DFT) calculations demonstrated that the synergistic modulation of thickness, pores, and N defects could remarkably increase the specific area, improve the light-harvesting capability, and enhance photoexcited charge carrier separation efficiency, strengthening the mass transfer in g-C_3N_4. As a consequence, the obtained DTLP-CN exhibits a wide spectrum response range and superior photocatalytic activity for H_2 production, with the highest hydrogen evolution rate (HER) of 1.557 mmol h^{-1} g^{-1} under visible-light irradiation, which is about 48.5 times higher than that of pristine g-C_3N_4. It is worth noting that the average apparent quantum yield (AQY) has the highest value up to 18.5% and reaches as high as 0.82% at 500 nm, exceeding most of the previously reported g-C_3N_4 photocatalysts. The study provides a useful strategy to design and fabricate highly efficient g-C_3N_4-based catalysts by synergistically regulating the different properties.

2. EXPERIMENTAL SECTION

2.1. Preparation of Photocatalysts. The bulk samples were prepared by a thermal polymerization process. In a typical synthesis, a certain amount of KOH was dissolved with stirring into 20 mL of deionized water; then, the mixture of precursors with 0.75 g U and 2.25 g M was dispersed into the above solution, followed by sonication, and stirring at ambient temperature for about 30 min and 1 h, respectively. The resulting suspensions were stirred and dried at 80 °C overnight to remove the solvent. After that, the white powders were transferred into a crucible and then calcined at 500 and 520 °C sequentially for 2 h in a muffle furnace with a ramping rate of 2 °C min^{-1}. The obtained solids were collected and ground into powders, followed by washing with deionized H_2O several times until the residual KOH was removed, and then the products were dried at 60 °C overnight. According to the procedure mentioned above, samples with different KOH contents were synthesized and noted as *m*KCN (where *m* = 1, 5, 10, 15, 20, 40, 100, and 200 mg is the mass of KOH used in the experiment). For comparison, samples without U or KOH in the precursor were prepared by the same way and labeled as *m*KMCN-B and M_xU_y-B (x:y = 3:1, 1:1, 1:2, 1:3, and 1:4), respectively. Similarly, MCN-B and UCN-B were also prepared using 3 g M and 3 g U, respectively.

The thin-layered porous g-C_3N_4 was prepared by a thermal oxidation etching process as follows: the obtained powders were uniformly dispersed into an alumina porcelain boat and calcined at 500 °C for 2 h in an open system. The final samples were denoted as *m*KCN, *m*KMCN, M_xU_y, MCN, and UCN, respectively. However, no UCN was obtained from the same preparation procedure.

2.2. Photocatalyst Characterizations. The morphology of as-prepared photocatalysts was characterized by a field-emission scanning electron microscope (Nano SEM650) and high-resolution transmission electron microscopy (HRTEM, Tecnai G2 F30 S-TWIN). The crystalline structure of the catalysts was examined by the X-ray diffraction (XRD) technology on D/max-2500/PC with a Cu Kα radiation source. The thicknesses of the catalysts were investigated by atomic force microscopy (AFM, Bruker Dimension Edge). Fourier-transform infrared (FTIR) spectra were collected on a Vertex 70 Fourier-transform infrared spectrophotometer (Bruker, German) over a range of 4000−400 cm^{-1} at a resolution of 2 cm^{-1}. Nitrogen adsorption and desorption isotherms were recorded on a Quadrasorb SI-MP system. The specific surface and pore-size distributions of the samples were calculated by the Brunauer−Emmett−Teller (BET) method and the Barrett−Joyner−Halenda (BJH) model, respectively. Solid-state ^{13}C magic angle spinning (MAS) NMR spectra were collected on a Bruker AVANCE III HD 500 MHz WB solid-state NMR spectrometer at room temperature. The C and N contents of samples were obtained by a Vario EL cube elemental analyzer. X-ray photoelectron spectroscopy (XPS) measurements were performed on a Thermo VG ESCALAB 250 spectrometer with Al Kα radiation as the excitation source. The UV−vis diffuse reflectance spectra (DRS) were recorded on a TU1950 spectrophotometer fitted with an integrating sphere attachment with the $BaSO_4$ powder as the reference. Steady-state photoluminescence (PL) spectrum measurements were detected by a FlouroMax-4 fluorescence spectrophotometer under an excitation wavelength of 310 nm. The time-resolved photoluminescence (TRPL) spectra were analyzed on a spectrometer from Edinburgh Analytical Instruments (F900) with a resolution of 100 ps. The electron paramagnetic resonance (EPR) measurements were performed on a Bruker (Bruker E500) EMX X-band spectrometer (microwave frequency = 9.8 GHz) at room temperature with a 300 W Xe lamp (PLS-SXE300D, Beijing PerfectLight) as the light source. All samples were measured under the same conditions (microwave power: 10 mW, receiver gain: 30 dB, modulation frequency: 100 kHz, modulation amplitude: 2 G, sweep time: 82 s).

2.3. Photoelectrochemical Measurements. To investigate the photoelectrode responses of the obtained samples, the PEC measurements such as transient photocurrent response under a fixed voltage and electrochemical impedance spectroscopy (EIS) were performed in a standard three-electrode system with a 0.1 M Na_2SO_4 solution on an electrochemical workstation (CHI 660E, Inc., Shanghai). A Pt foil and Ag/AgCl (saturated KCl solution) electrodes were used as the counter electrode and the reference electrode, respectively. A 300 W Xe lamp (PLS-SXE300/300UV) with an ultraviolet cutoff filter (λ > 420 nm) was used to obtain the incident light. All working electrodes were fabricated according to the following procedures: 10 mg of samples was dispersed into a mixture of 1 mL of ethanol and 0.1 mL of Nafion. After continuous ultrasonic treatment for 1 h, the obtained slurry was spin-coated onto a 1.5 cm × 1.5 cm fluorine-doped tin oxide (FTO) glass and then annealed at 80 °C for 10 h to enhance adhesion. Transient photocurrent response was recorded at −0.4 V versus Ag/AgCl electrodes with a chopper controlling the time intervals of light on and off. Electrochemical impedance spectroscopy (EIS) was measured at an amplitude of 5 mV with a frequency range from 10^5 to 10^{-2} Hz.

2.4. Theoretical Calculation Details. The DFT calculations in this work were performed by employing the Vienna Ab initio Simulation Package (VASP).[42,43] The interaction between the core and valence electrons is described using the frozen-core projector-augmented wave approach (PAW).[44] The Perdew−Burke−Ernzerhof of generalized gradient approximation (PBE-GGA) was chosen to describe the exchange−correlation interaction. A plane-wave energy cutoff of 450 eV was set. The lattice constants and all atomic positions were fully relaxed by the conjugate gradient procedure and the tolerance for energy convergence was set to 10^{-5} eV. A Monkhorst−Pack mesh of 4 × 4 × 1 *k*-points was used to sample the Brillouin zone of the unit cell of g-C_3N_4 and the defected g-C_3N_4.[4545] The

https://dx.doi.org/10.1021/acsami.0c14262
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

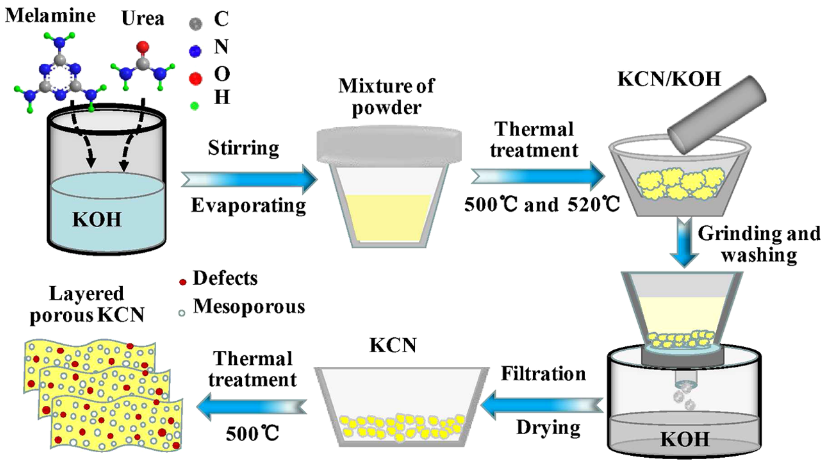


Figure 1. Schematic illustration for the preparation of layered and porous KCN using the mixture of melamine, urea, and KOH as precursors.

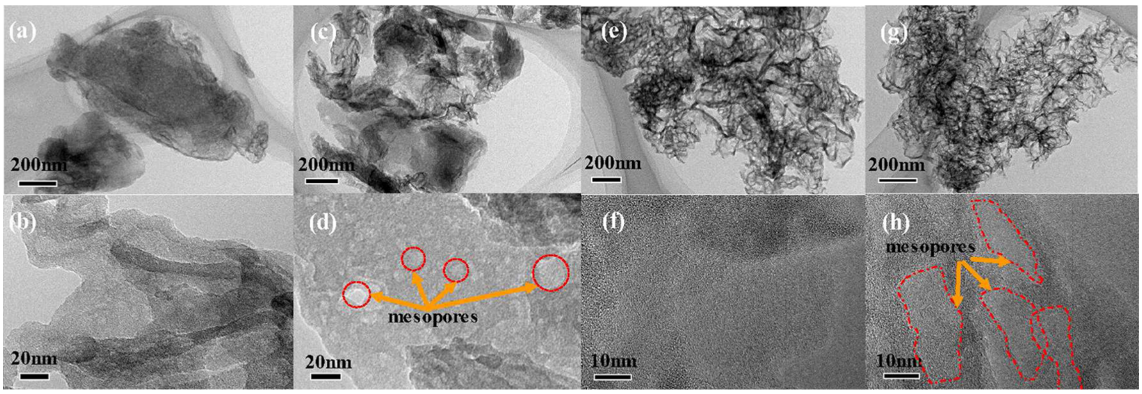


Figure 2. TEM images of MCN (a, b), 10KMCN (c, d), M_1U_3 (e, f), and 10KCN (g, h).

periodic boundary condition was applied in the two-dimensional (2D) plane, while a 15 Å vacuum space was inserted in the out-of-plane direction to avoid the interaction between each periodic image. The relaxed lattice constants are a = 16.81 Å and b = 12.82 Å.

2.5. Photocatalytic H_2 Evolution Measurements. All of the photocatalytic H_2 evolution tests were performed on an online automatic photocatalytic test system (Labsolar-IIIAG, Beijing Perfect-light) equipped with a Shimadzu GC-2014 gas chromatograph. Typically, 50 mg of the photocatalyst was dispersed in 100 mL of an aqueous solution containing 20% triethanolamine (TEOA) as the sacrificial agent for hydrogen evolution. For all reactions, 3 wt % of Pt as a cocatalyst was in situ photodeposited on the surface of photocatalysts by dropping a certain amount of a H_2PtCl_6 solution as a Pt precursor. Prior to light irradiation, the suspension was evacuated by a vacuum pump and continuously stirred by a magnetic stirrer, and then a 500 W xenon lamp equipped with a 420 nm cutoff filter was used as a light source for the H_2 evolution reaction. During the reaction process, the reaction temperature was carefully maintained at 5 °C by a low-constant temperature bath (DC-2006, Shanghai Bilang). The amount of evolved hydrogen was recorded and analyzed by the online GC (argon carrier gas, thermal conductivity detector (TCD), and flame ionization detector (FID), molecular sieve 5 Å). To detect the roles of Pt and TEOA during the photocatalytic reactions, the photocatalytic H_2 evolution measurements without Pt or TEOA were also performed. Moreover, the aqueous solution after the photocatalytic reaction was characterized by high-performance liquid chromatography (HPLC) and liquid chromatography−mass spectrometry (LC/MS) measurements. In addition, according to a similar procedure as above, the apparent quantum efficiency (AQE) for H_2 evolution was detected under the irradiation of a monochromatic light with different wavelengths. The irradiation intensity was recorded by a radiation meter (Thorlabs, PM100D with S310C). The irradiation area was controlled as 6.25 cm^2. The AQE was calculated by the following equation

$$\mathrm{AQE} = \frac{N_e}{N_i} \times 100\% = \frac{2 \times R \times N_a \times h \times c}{P \times S \times t \times \lambda} \times 100\%$$

where N_e and N_i are the number of electrons and incident photons, respectively, R is the amount of hydrogen, N_a is the Avogadro constant, h is the Planck constant, c is the speed of light, P is the intensity of the irradiation light, S is the irradiation area, t is the time of the photoreaction, and λ is the wavelength of the monochromatic light. For a stability test, the as-prepared sample was used to catalyze water splitting for the first 5 h, according to the method mentioned above. Then, the reaction system was vacuumized to evacuate the generated H_2 completely. After that, the photocatalytic reaction was continued for another 5 h. The same process was conducted four times.

3. RESULTS AND DISCUSSION

The process of preparing DTLP-CN is shown in Figure 1. The bulk samples were synthesized via simply calcining the mixture of M, U, and KOH in proper proportions. After grounding and washing with H_2O to remove the residual KOH and then annealing at 500 °C for 2 h in static air, the thin-layered and porous g-C_3N_4 samples were obtained.

The morphology and microstructure of all prepared samples were investigated by scanning electron microscopy (SEM) and transmission electron microscopy (TEM). As shown in Figure 2, MCN and 10KMCN are stacked by g-C_3N_4 sheets and possess a thick layered structure, while M_1U_3 and 10KCN show an ultrathin sheet-like structure, in stark contrast to the

https://dx.doi.org/10.1021/acsami.0c14262
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

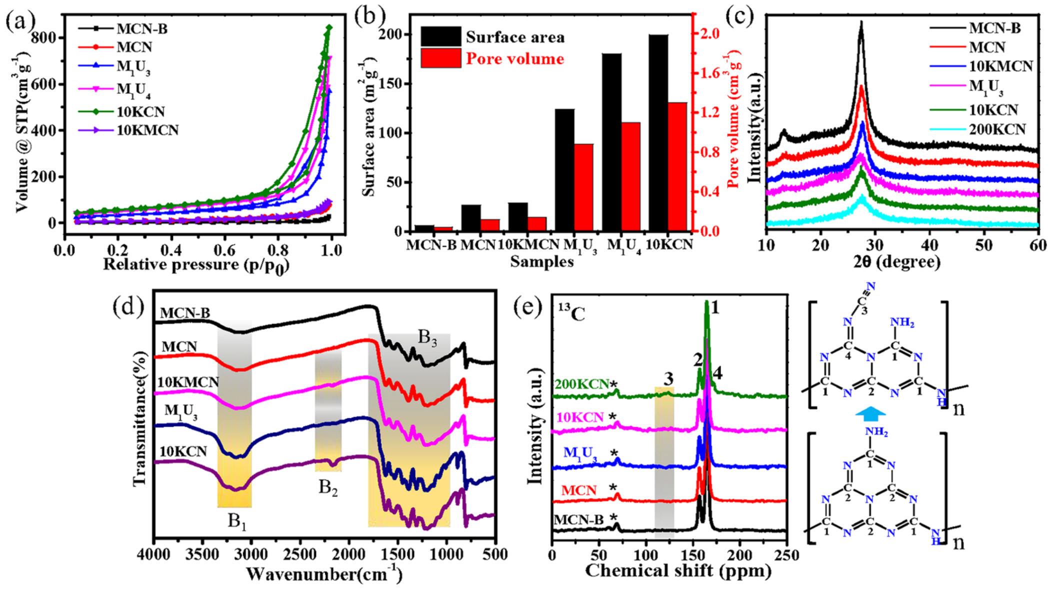


Figure 3. (a) Nitrogen adsorption−desorption isotherms, and the (b) surface areas and pore volumes of MCN-B, MCN, M_1U_3, M_1U_4, 10KCN, and 10KMCN. (c) XRD patterns of MCN-B, MCN, M_1U_3, 10KCN, 10KMCN, and 200KCN. (d) FTIR spectra of MCN-B, MCN, M_1U_3, 10KCN, and 10KMCN. (e) Solid-state ^{13}C magic angle spinning (MAS) NMR spectra of MCN-B, MCN, M_1U_3, 10KCN, and 200KCN. The spinning side bands are represented by "*" in the ^{13}C spectra.

bulk MCN and 10KMCN. The SEM images of the samples are presented in Figures S1 and S2, where MCN-B and MCN show a bulk structure. However, due to the thermal oxidation and polycondensation of U during the polymerization process, all of the M_xU_y samples show a flower-like nanosheet structure and get smoother and thinner with increasing the mass ratio of U, which could significantly improve the surface area of the sample and provide more exposed active sites for the photocatalytic reaction.[30,46] According to the AFM image (Figure S3), the thicknesses of MCN and 10KMCN are slightly decreased in comparison with MCN-B. However, they are apparently decreased to about 30 nm for M_1U_3 and 10KCN, and the layered structures are composed of some thin sheets with a thickness of around 10 nm, which suggests that the addition of U could facilitate the decrease of the thickness of the obtained samples. In addition, as shown in the high-resolution TEM images (Figures 2d,h and S4), numerous mesopores can be apparently observed on the surface of 10KMCN and 10KCN, while they cannot be observed on MCN and M_1U_3, mainly due to the etching effect of KOH during thermal treatment.[40,47,48] Meanwhile, the escape of CO_2, H_2O, and NO_2 during the thermal oxidation and polymerization process is also beneficial for the formation of pores on the surface of *m*KCN or *m*KMCN. The formed in-plane mesopores in g-C_3N_4 are beneficial for enhancing the mass-transfer process,[39,40] resulting in more charge carriers and reactants involved in the photocatalytic reactions, which is beneficial for improving the efficiency. Furthermore, the SEM images in Figure S2 indicate that all of the *m*KCN samples possess sheet-like structures. Thus, it can be concluded that the mesoporosity of g-C_3N_4 can be modulated by the addition of KOH, while the characteristic sheet structure of the samples is largely preserved.

To further investigate the effect of U and KOH addition on the specific surface area and porous structure of samples, the N_2 adsorption−desorption measurements were carried out at 77 K. As shown in Figure 3a, all of the samples show type-IV isotherms with characteristic hysteresis loops, indicating the existence of mesopores.[47−49] Meanwhile, the Barret−Joyner−Halenda (BJH) analysis (Figures 3b and S5, and Table S1) indicates that the M_1U_3, M_1U_4, and 10KCN are rich in pores compared with MCN and MCN-B. Especially, 10KCN and 10KMCN possess more abundant pores than M_1U_3 and MCN in the range from 10 to 70 nm, respectively, which is consistent with TEM results and further confirms the importance of KOH in the introduction of mesopores. Moreover, in comparison with MCN and 10KMCN, the Brunauer−Emmett−Teller (BET) specific surface areas of M_1U_3 and 10KCN improve dramatically with the addition of U, suggesting that the specific surface area of g-C_3N_4 can be facilely modified by changing the mass ratio of U. However, the biggest average pore (17.3 nm) of M_1U_4 suggests that the nanosheet structure would be partially collapsed if excessive U is added.[46] The collapsed structure can destroy the mesopores and decrease the active sites of the samples, which would be responsible for the catalyst deactivation of M_1U_4 to some extent. In addition, combined with the synergetic effect of U and KOH, 10KCN with the superb photocatalytic activity has a specific surface area up to 199.4 $m^2\ g^{-1}$ and a pore volume of 1.3 $cm^3\ g^{-1}$, which is 36.2 and 30.4 times higher than that of MCN-B, respectively, and is also much higher than that of 10KMCN and M_1U_3. This suggests that the UKATP strategy is more effective for modulating the properties of g-C_3N_4 compared to the method with the presence of only U or KOH. Thus, the thin-layered porous g-C_3N_4 can be facilely synthesized by the UKATP strategy, which can significantly enlarge the specific surface area of g-C_3N_4 and shorten the diffusion distance of photoexcited carriers, resulting in more efficient energy capture and more active sites for the photocatalytic reaction.

The structures of the obtained samples were characterized by X-ray diffraction (XRD) patterns and Fourier-transform infrared (FTIR) spectroscopy. As shown in Figure 3c, two characteristic diffraction peaks can be observed in the XRD patterns of the obtained samples. The weak peak at 13.2° is

https://dx.doi.org/10.1021/acsami.0c14262
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

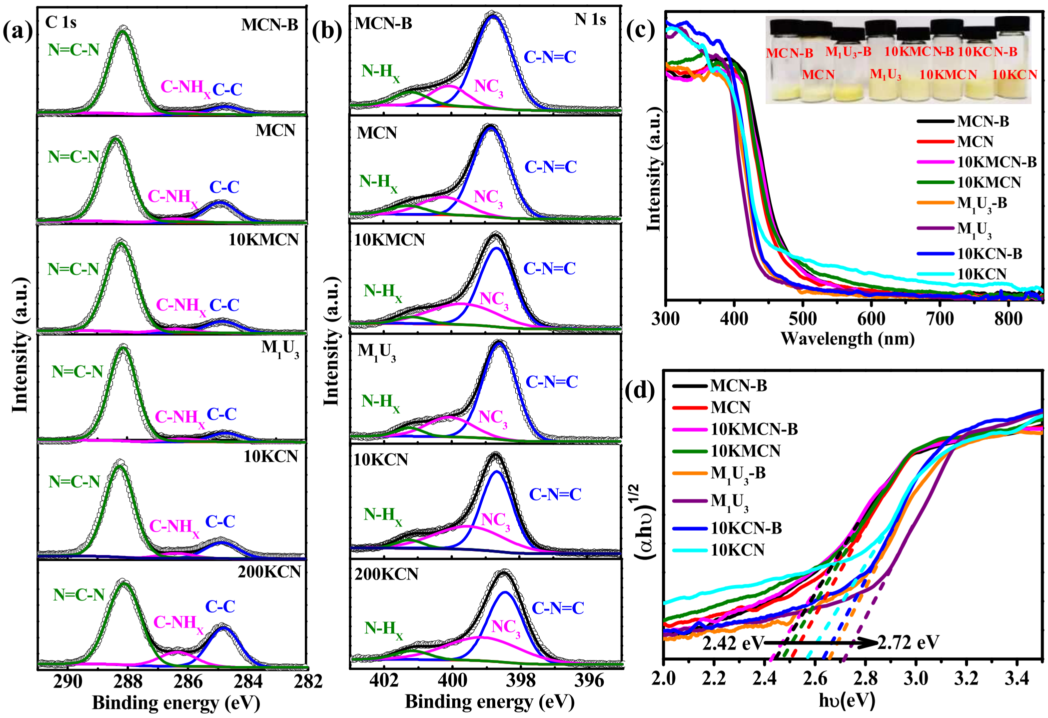


Figure 4. High-resolution XPS spectra of (a) C 1s and (b) N 1s in MCN-B, MCN, 10KMCN, M_1U_3, 10KCN, and 200KCN. (c) UV−vis diffuse reflectance spectra and (d) plots of $(\alpha h\nu)^{1/2}$ versus E_g (eV) of MCN-B, MCN, 10KMCN-B, 10KMCN, M_1U_3-B, 10KCN-B, and 10KCN.

related to the in-plane packing of tri-*s*-triazine units, indexed to the (100) crystal plane of g-C_3N_4, while the strong peak at 27.4° indexed to the (200) crystal plane is ascribed to interplanar stacking of the conjugated aromatic units.[10] As shown in Figure S6, the peak intensities of MCN and M_xU_y become gradually weaker with increasing the mass ratio of U, indicating a gradually thinner thickness of the g-C_3N_4 nanosheets, which further confirms that the thermal oxidation and polycondensation of U during the thermal treatment can effectively exfoliate the bulk g-C_3N_4 into thin nanosheets through overcoming the weak van der Waals force between layers. Furthermore, the two peaks of *m*KCN are slightly weakened and broadened with increasing usage of KOH; especially the peaks (100) located around 13° of 100KCN and 200KCN almost disappeared, implying that g-C_3N_4 or its precursors can react with KOH during the thermal treatment process, and excessive KOH addition can embed numerous defects into the in-planes of samples by reducing the orderliness of the structure within the framework.

The chemical structures of the obtained samples were further disclosed by the FTIR technology. As shown in Figure 3d, all of the bands corresponding to typical g-C_3N_4 are observed, suggesting that the chemical structures of most tri-*s*-triazine ring units are conserved after the thermal treatment. Specifically, the peaks at 810 cm^{-1} are assigned to the out-of-plane bending mode of heptazine rings, while the intense bands located in the region of 900−1800 cm^{-1} are attributed to the N−C═N heterorings in the "melon" framework. The broad peaks in the 3000−3500 cm^{-1} region are assigned to stretching vibrations of N−H and O−H generated from the amine group and adsorbed water.[50] Compared with MCN-B and MCN, the intensities of peaks of M_xU_y (Figure S7a) are stronger and enhanced gradually with increasing the mass ratio of U, suggesting more vigorous vibrations of atoms in the thin-layered g-C_3N_4, mainly due to the less restraint and higher exposure of atoms in two adjacent layers and tri-*s*-triazine rings caused by the exfoliation effect of U during thermal treatment.[39,46] Moreover, for the *m*KMCN and *m*KCN samples (Figure S7b,c), two obvious differences can be found in the spectra with increasing addition of KOH. The first is the gradually decreased intensity of the N−H stretching vibration between 3000 and 3300 cm^{-1}. The second is the emergence of a new band at 2178 cm^{-1}, ascribing to the asymmetric stretching vibration of C≡N triple bonds from cyano groups. All of these suggest that the cyano groups can be introduced and may originate from the deprotonation of −C−NH_2 groups with the addition of KOH, which is mainly due to that the OH^- derived from the melted KOH can react with the amine groups during the thermal treatment process.[19,51] In addition, although with the same mass of KOH in the precursors, the peak of cyano groups for *m*KCN is stronger than that of *m*KMCN, such as 10KCN and 10KMCN (Figures 3d and S7), suggesting that more cyano groups are generated in *m*KCN and the addition of U is beneficial for their introduction. This could be ascribed to the polycondensation and the exfoliation effect of U, which can make the melted KOH contact more fully and react well with the amine groups during the thermal treatment process. Thus, it can be concluded that the thickness and the concentration of cyano groups of g-C_3N_4 can be facilely regulated by changing the composition of the precursor before the thermal polymerization process.

Solid-state and ^{13}C cross-polarization magic angle spinning nuclear magnetic resonance (CP-MAS-NMR) measurements were conducted to further identify the molecular structures of the synthesized samples. As shown in Figure 3e, two high peaks at 156.3 and 164.5 ppm can be observed for all samples, corresponding to the characteristic chemical shift of C_{3N} (2) atoms and C_{2N-NH_x} (1) atoms in heptazine, respectively.[19,25,52,53] However, as shown in Table S2, the intensity

https://dx.doi.org/10.1021/acsami.0c14262
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

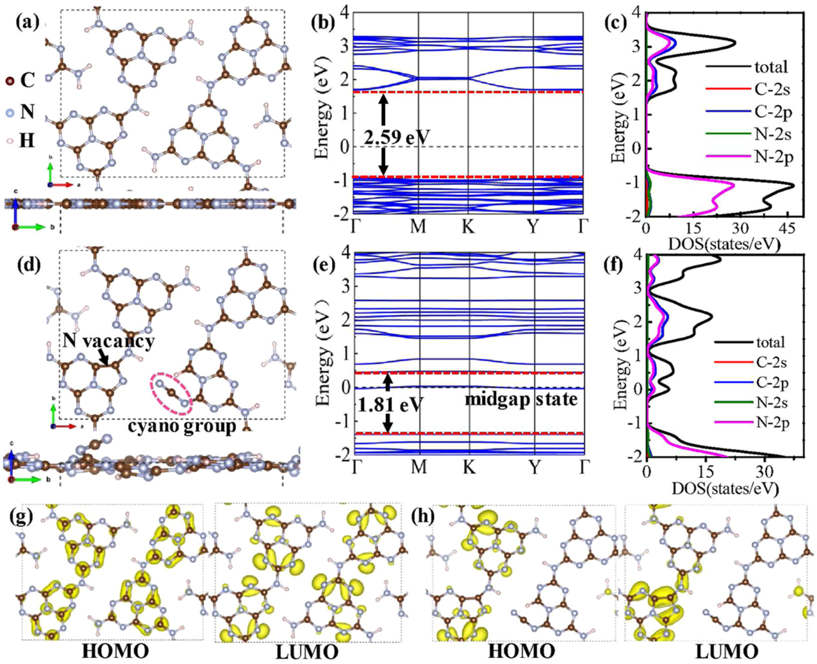


Figure 5. Structure models (a, d), calculated band structures (b, e), and corresponding DOS (c, f) of pristine g-C_3N_4 and *m*KCN with N defects, respectively. The charge density distribution of the highest occupied molecular orbital (HOMO) and LUMO for pristine g-C_3N_4 (g) and *m*KCN (h).

of peak at 156.3 ppm is progressively decreased with increasing the use of KOH, and the intensity ratio of C_{3N} to C_{2N-NH_x} decreases from 0.29 in MCN-B to 0.14 in 200KCN. Moreover, two new peaks at 123.3 and 171.4 ppm are present in 200KCN, which can be assigned to the C (3) atoms in cyano groups and the neighboring C (4) atoms, respectively. Based on the ^{13}C NMR results, it can be concluded that the introduced cyano groups stemmed from the deprotonation of $-C-NH_2$ groups may be located at the apex of the melon structure of *m*KCN, which can lead to partial fracturing and restructuring of $N{=}C-N_2$ units, and ultimately result in the decreased concentration of C_{3N} (2) atoms.

The effects of U and KOH addition on the structure and elemental composition of the synthesized samples were further detected by the X-ray photoelectron spectroscopy (XPS) and organic elemental analysis (OEA) measurements. As shown in Table S3, the N/C atomic ratios of MCN-B, MCN, and M_1U_3 obtained from the above two methods are close to the theoretical value of g-C_3N_4,[53] with the exception of a slight decrease for M_1U_3 obtained from XPS, which suggests that the addition of U during the thermal treatment might just introduce little N defects on the surface and does not change the whole elemental composition of M_xU_y. According to the above FTIR analysis, the introduction of cyano groups does not lead to N reduction, while the N/C ratio is decreased with increasing addition of KOH, and the XPS data shows a faster reduction, implying that much more N vacancies are introduced and most of them are on the surface of *m*KCN. The rich N vacancies in g-C_3N_4 are beneficial for the formation of a large number of two neighbor two-coordinated C atoms (C_{2C}), which can act as electron traps. Moreover, the N vacancies can greatly enhance the reactive metal−support interaction, which not only facilitates to form the high coverage density and stability of Pt but also can promote the ability of Pt for capturing the photoexcited electrons, resulting in the high separation and transfer efficiency of charge carriers in g-C_3N_4.[54]

Such findings can be further corroborated by the XPS survey and their high-resolution spectra (Figure 4a,b). The narrow scan C 1s XPS spectra of the samples can be deconvoluted into three peaks located at 284.8, 286.3, and 288.3 eV, respectively, which are attributed to carbon in adventitious hydrocarbons (C−C), sp^2-bonded C in the aromatic ring attached to $-NH_x$ (x = 1, 2) on the edges of heptazine units ($C-NH_x$), and sp^2-hybridized C in the N-containing aromatic ring of g-C_3N_4 ($N-C{=}N$), respectively.[15,19,39] Obviously, the peak at 286.3 eV is progressively intensified with increasing KOH usage, and the peak area ratio of $C-NH_x/C_{C1s}$ is increased from 0.033 to 0.116 (Table S4), which could be ascribed to the formation of cyano groups during the polycondensation process because the binding energy of C 1s in cyano groups is similar with that in the $C-NH_x$.[51] Furthermore, as presented in Figure 4b, the N 1s XPS spectra for MCN-B, MCN, and M_1U_3could be fitted into three peaks centered at 398.7, 400.2, and 401.2 eV, which are assigned to N atoms in the $C-N{=}C$ (N_{2C}), NC_3, and NH_x groups of melon, respectively.[20,55] Interestingly, compared with MCN and M_1U_3, the NC_3 peaks in 10KMCN, 10KCN, and 200KCN spectra show a slight shift to the lower binding energy, suggesting the introduction of cyano groups, since the N 1s binding energy of the cyano group is intermediate between that of N_{2C} and NC_3.[19] Moreover, as shown in Figure S8 and Table S5, the peak area ratio of N_{2C}/N_{N1s} indicates an apparent decrease with increasing KOH usage, dropping from 0.744 to 0.548, which could be ascribed to the generation of N vacancies on the surface of *m*KCN. Additionally, in comparison with MCN-B, the peak area ratios of $C-NH_x/C_{C1s}$ and N_{2C}/N_{N1S} for 10KCN-B show a larger change than that for 10KMCN-B, and the differentials are slightly increased after the second thermal treatment, suggesting that the exfoliation effect of the second thermal

https://dx.doi.org/10.1021/acsami.0c14262
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

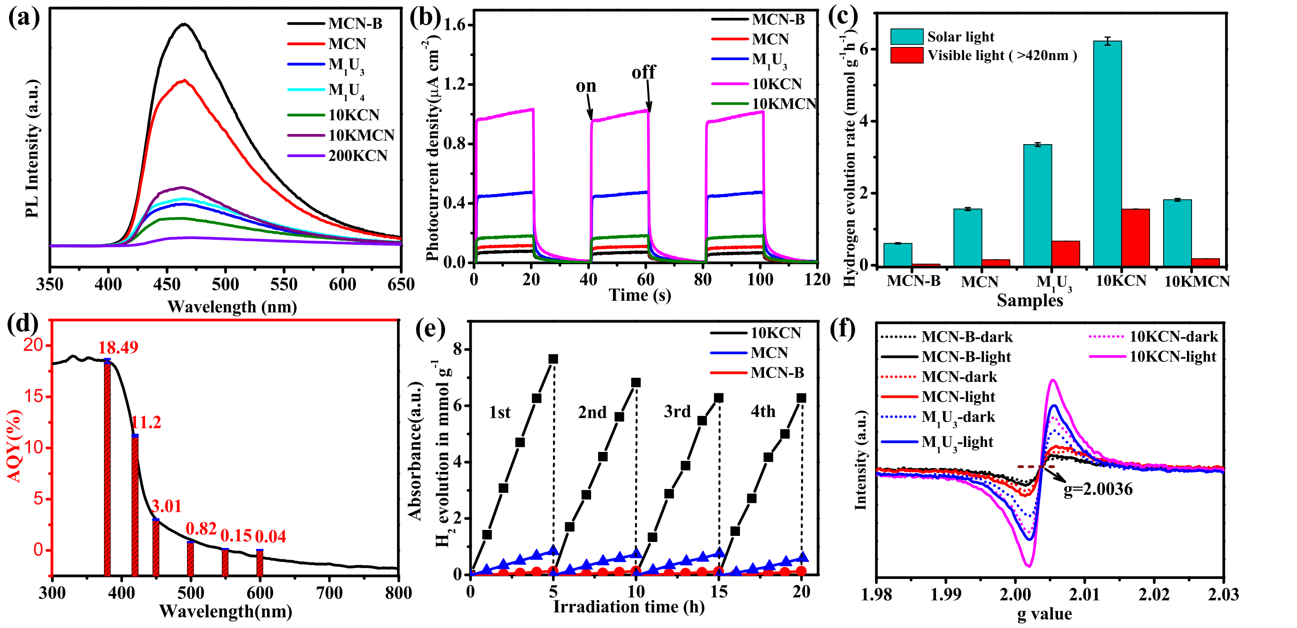


Figure 6. (a) Photoluminescence spectra for MCN-B, MCN, M_1U_3, M_1U_4, 10KCN, 10KMCN, and 200KCN. (b) Transient photocurrent response of MCN-B, MCN, M_1U_3, 10KCN, and 10KMCN under visible-light illumination (λ > 420 nm). (c) Photocatalytic H_2 evolution rates of MCN-B, MCN, M_1U_3, 10KCN, and 10KMCN. (d) Absorption spectrum and wavelength-dependent AQY of photocatalytic H_2 evolution for 10KCN. (e) Cycling test of H_2 evolution curves for MCN-B, MCN, and 10KCN. (f) EPR spectra of MCN-B, MCN, M_1U_3, and 10KCN with or without illumination.

treatment is beneficial for exposing more N defects on the surface of g-C_3N_4. These further confirm that the UKATP strategy is more effective for introducing and regulating the N defects in g-C_3N_4.

The optical properties of the obtained samples are investigated by UV−visible diffuse reflectance spectra (DRS). As shown in Figures 4c,d and S9a,d, a progressive blue shift in the absorption edge is achieved with increasing the mass ratio of U. Correspondingly, the band gaps of the samples calculated from Kubelka−Munk plots are progressively broadened from 2.44 to 2.73 eV. This result can be taken as additional evidence for the effective exfoliation role of U during the thermal polycondensation process because the reduction in thickness would broaden the band gap of semiconductors.[30,56] Moreover, in comparison with MCN and M_1U_3, the absorption edges of *m*KMCN and *m*KCN show a progressive red shift with increasing the KOH usage (Figures 4c,d and S9b−f), demonstrating that their band gaps are progressively narrowed. Meanwhile, their absorption intensities are obviously enhanced in the whole visible-light region, implying that the defect-related midgap state is introduced and reforms the band structure of g-C_3N_4.[25] Additionally, with adding the same mass of KOH into the precursor of MCN and M_1U_3, the spectral responsivity and band gap show a more dramatic change for the resulting 10KCN-B in comparison with M_1U_3-B than that for 10KMCN-B in comparison with MCN-B, and the differentials are further increased for 10KCN and 10KMCN in comparison with M_1U_3 and MCN, respectively (Figure 4c,d). Based on the XPS results, this is mainly due to the more efficient introduction of N defects by the UKATP strategy, and more N defects are exposed on the surface of the g-C_3N_4 after the second thermal treatment, which can directly affect the band structures and optical properties of the samples. As determined from the valence band (VB) XPS spectra (Figure S10a), the VBs are similar for the six samples. Combined with the obtained band gap values, the band structure alignments are shown in Figure S10b. It can be concluded that the thermal exfoliation and introduction of N defects mainly change the conduction band (CB) of the synthesized g-C_3N_4. All of these results give evidence of the availability of the UKATP strategy for modulating the band structure and the corresponding light-harvesting capability of g-C_3N_4 well.

To further detect the change in the band gap and light-harvesting capability of g-C_3N_4 caused by the introduced N defects, the electronic structures of g-C_3N_4 and *m*KCN were calculated by first-principles based on DFT. As shown in Figure 5a−f, a direct band gap for the pristine g-C_3N_4 is 2.59 eV. The density of states (DOS) calculations indicate that the VB is mainly composed of N 2p orbitals while both the N 2p and C 2p orbitals contribute to the CB of g-C_3N_4, which is consistent with the previous results.[19,25] With the introduction of the cyano group and a N vacancy into the unit cell of g-C_3N_4, the band gap energy decreases to 1.81 eV and the localized states are created at a range of 0.5−1.0 eV below the lowest unoccupied molecular orbital (LUMO). Generally, a high density of localized states is closely related to the band tails, which can form a continuum extension to and overlap with the LUMO rather than forming discrete donor states below the LUMO.[20,57] This could be the crucial reason for the *m*KCN and *m*KMCN with the narrowed band gap and an extended optical response range. Furthermore, a midgap energy level ascribed to N 2p and C 2p orbitals is generated above the VB, which is consistent with the enhanced absorption intensities for the *m*KCN and *m*KMCN in the visible-light region observed in the DRS results. Additionally, to further investigate the contributions of the N vacancy and the cyano group toward the final band structure, the models with different types of N defects were constructed separately. The calculations in Figure S11 show that: (i) compared with pristine g-C_3N_4, the CB minimum is lowered by 0.76 eV after

https://dx.doi.org/10.1021/acsami.0c14262
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

the introduction of a cyano group, (ii) a midgap energy level appears about 1.36 eV above the VB, mainly attributed to the defect state originated from the introduced N vacancy. These results demonstrate that the coexistence of cyano groups and N vacancies are the key to modulate the band structure of g-C_3N_4.

In addition, the result orbitals of pristine g-C_3N_4 are shown in Figure 5g, where the spatial distributions of charge density of LUMO and HOMO are uniformly delocalized over the heptazine rings. In contrast, as shown in Figure 5h, the charge density in *m*KCN is redistributed with the appearance of some electron-rich regions, and the distributions corresponding to the LUMO and HOMO are separated well, which may be attributed to the enhanced local fluctuations of the electrostatic potential caused by the introduced N defects. Such separated and localized distribution of charge density would keep the redox sites separated from each other, facilitating the separation and transfer of photoexcited charge carriers, which can also restrain the back reactions and improve the photocatalytic activity of g-C_3N_4 significantly. More importantly, these results suggest that the introduction of N defects into g-C_3N_4 by the KOH-assisted thermal polymerization with the exfoliation simultaneously could be a very effective strategy for modulating the band gap and enhancing the charge separation.

The separation and transfer properties of photoexcited charge carriers were characterized by photoluminescence (PL) spectroscopy. As presented in Figures 6a and S12, in comparison with the MCN-B, the PL intensities of the samples with U- and KOH-assisted thermal treatment are weakened progressively, confirming that the charge recombination can be inhibited with the thermal exfoliation and introduction of N defects simultaneously. Meanwhile, the corresponding time-resolved fluorescence spectra of the samples were recorded to investigate the average radiative lifetime (τ_{ave}) of the photoexcited charge carriers (Figure S13). The calculated τ_{ave} (Table S6) of MCN-B is 9.657 ns, and τ_{ave} is increased with the addition of U during the thermal polymerization process, suggesting the enhanced exciton dissociation can be achieved by this thermal exfoliation process, and the suppressed recombination processes and longer diffusion length of charge carriers are further confirmed. For M_1U_4, the shorter τ_{ave} than that of M_1U_3 is probably due to the excessive defects originated from its collapsed structure, which could generate the deeper midgap states and act as recombination centers,[46,58] resulting in the enhanced charge recombination and stronger PL intensity compared with M_1U_3. In addition, τ_{ave} is gradually decreased with addition of KOH and eventually reduced to 3.191 ns when 200 mg of KOH is used, further confirming that the introduced N defects can enhance the separation and transfer of photoexcited charge carriers substantially.

To further understand the separation and transport property of the photoexcited charge carriers, photoelectrochemical measurements are conducted. As shown in Figure 6b, all of the obtained samples demonstrate sensitive photocurrent response during several on−off cycles of intermittently visible-light irradiation. Compared with MCN-B, the photocurrents of MCN and M_1U_3 are increased apparently, suggesting that the increased surface area and the corresponding thinner thickness caused by the thermal treatment are beneficial for generating more charge carriers and suppressing their recombination. Moreover, the photocurrent of 10KMCN is higher than that of MCN, and 10KCN has the highest photocurrent, which is mainly due to the introduction of mesopores and N defects simultaneously into g-C_3N_4 by U- and KOH-assisted thermal treatment, suggesting that the thin-layered and porous structure makes a better visible-light response and more efficient photoexcited charge-separation property for photocatalytic reactions. The above results were further clarified by electrochemical impedance spectroscopy (EIS). As shown in Figure S14, the EIS Nyquist plot of 10KCN recorded under visible-light irradiation indicates a smaller arc radius than that of other samples, demonstrating the high conductivity and the corresponding easier and faster charge transfer in 10KCN.

The photocatalytic activities of all samples were evaluated with H_2 evolution from the mixture of water and triethanolamine (TEOA) under visible-light and solar-light irradiations, respectively. As shown in Figures 6c and S15−S17, all samples with a 3 wt % Pt cocatalyst show the photocatalytic H_2 evolution, and no obvious signal of the H_2 peak appeared in the gas chromatogram (GC) (Figure S17a,b), demonstrating the necessity of the coexistence of a cocatalyst and a hole sacrificial agent. The increase for MCN compared with MCN-B is owing to the exfoliation effect of the second thermal treatment. Moreover, the photocatalytic performance is progressively enhanced with decreasing the mass ratio of M to U from 3:1 to 1:3 (Figure S16a,b), mainly attributed to that the thermal treatment with U can reduce the thickness and increase the specific surface area of g-C_3N_4 significantly. Whereas, the photocatalytic performance of M_1U_4 begins to decrease, which is primarily due to the less active sites and lots of deeper midgap states, arising from its collapsed nanosheet structure. Additionally, 10KMCN shows a higher HER than that for MCN, suggesting that the photocatalytic activity can be enhanced by the N defects and mesopores introduced by the addition of KOH. Thus, as shown in Figure S16c,d, the photocatalytic activity of *m*KCN is notably improved on account of the synergistic effects of U and KOH. Especially, 10KCN presents the highest photocatalytic HER of 1.558 mmol g^{-1} h^{-1} under visible-light irradiation, which is about 48.5 times higher than that of MCN-B (0.032 mmol g^{-1} h^{-1}), while the catalytic rate for M_1U_3 and 10KMCN are only about 20.9 and 5.7 times higher than that of MCN-B, respectively. This fully demonstrates that modulating and optimizing the diverse parameters of g-C_3N_4 synergistically can boost the catalytic activity maximally. The superior enhancing effect of this U- and KOH-assisted method is larger than most of the previous reports and makes it one of the optimal strategies for enhancing the photocatalytic activity of g-C_3N_4 (Table S7). The superior property of 10KCN can be attributed to its significantly enhanced light-absorption capacity, better charge-separation efficiency, and rapid mass-transfer process caused by its thin-layered and porous structure. However, for *m*KCN, the activity is gradually decreased with increasing KOH usage (>10 mg), which is mainly due to that the excessive N defects and pores can lower the CB position and introduce lots of deeper midgap states, resulting in decreased reaction kinetics and less photogenerated electrons for hydrogen evolution. The above results imply that there might be an optimum level of pores and N defects in *m*KCN for achieving the highest HER. Moreover, the AQY of 10KCN shows the highest value of 18.5% at 380 nm, and it reaches 11.2% at 420 nm and 0.87% at 500 nm, which is higher than many previously reported studies (Table S7), indicating its advantage as a visible-light-

https://dx.doi.org/10.1021/acsami.0c14262
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

responsive photocatalyst. Additionally, the catalytic stability of 10KCN was investigated by cycling experiments. As shown in Figure 6e, the photocatalytic activity was sustained after four cycles. No significant variation in photocatalytic HER and morphology (Figures S18 and S19) could be detected, and the HPLC and LC/MS measurements indicate that the TEOA is oxidized to DEOA and glycolaldehyde by the photogenerated holes during the photocatalytic reactions (Figure S20). These are consistent with the previous reports and demonstrate the high stability of 10KCN as a photocatalyst.[59,60]

For in-depth understanding of the photocatalytic mechanism, the photoexcitation and electron transfer processes of the obtained samples were explored by the room-temperature electron paramagnetic resonance (EPR) spectra. As illustrated in Figure 6f, all samples show one single Lorentzian line with a g value of 2.0036, which is induced by surface trapped electrons that preferentially arise from sp^2-carbon atoms within the π-conjugated aromatic rings.[61,62] In the dark situation, the EPR signal is gradually enhanced with addition of U and KOH, and 10KCN possesses the strongest intensity. As the intensity of EPR is positively correlated to the concentration of unpaired electrons,[25,34] it can be concluded that the U- and KOH-assisted thermal polymerization can decrease the thickness and introduce more mesopores and N defects into g-C_3N_4 clearly, which would form more unsaturated sites in g-C_3N_4. This is beneficial for electron excitation from the VB to the unoccupied orbitals of CB and forming more unpaired electrons. As expected, the thin-layered and porous 10KCN exhibits the highest EPR signal under visible-light irradiation, suggesting more photoexcited electrons are trapped in CB, which can quickly transfer to the Pt nanoparticles on the surface of 10KCN, resulting in its optimally photocatalytic activity for H_2 evolution.

On the basis of the experimental and calculated results, the mechanism of the optimal photocatalytic activity of the 10KCN for H_2 evolution can be proposed. As shown in Figure 7, 10KCN prepared by the UKATP strategy possesses a

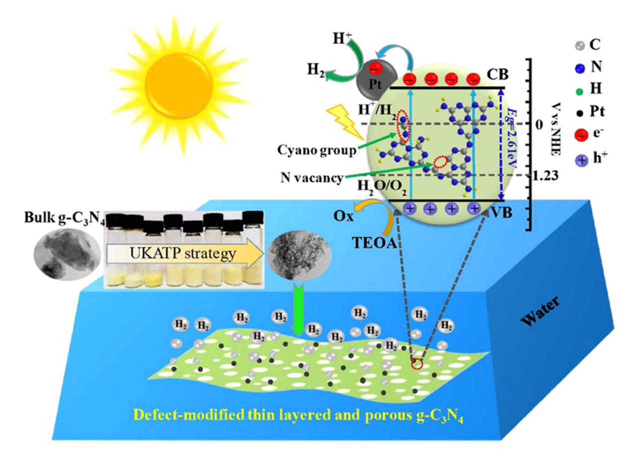


Figure 7. Photocatalytic H_2 evolution mechanism of defect-modified thin-layered and porous g-C_3N_4.

higher specific surface area, favoring the maximized exposure of the surface to the incident light and surface active sites to the reactant. Moreover, the defect-modified thin-layered and porous structure of 10KCN cannot only minimize the distance that the photogenerated carriers need to migrate to the surface active sites significantly but also can introduce plentiful of unsaturated sites in g-C_3N_4 and generate more unpaired electrons, which can improve the generation rate of photoexcited electrons dramatically, facilitating their separation and migration to the Pt nanoparticles on the surface of 10KCN. Besides, the formed in-plane mesopores in 10KCN are more favorable for the rapid mass-transfer process, and the corresponding cross-plane diffusion of photoexcited carriers, reactant, and hydrogen can be achieved facilely, which can improve the reaction kinetics and accelerate the photocatalytic reaction largely. In addition, the increased density of band tail states/localized states derived from the introduced N defects can lower the CB position, leading to 10KCN with a narrower band gap and higher light-harvesting capability. The defects can also act as separation centers to trap photogenerated electrons from the conduction band of 10KCN and improve the efficiency of carrier separation.[35] As a result, the photocatalytic activity of 10KCN for H_2 evolution is dramatically improved due to the synergetic enhancement effects derived from the UKATP strategy, which simultaneously modulate the morphological, structural, optical, and electronic properties.

4. CONCLUSIONS

In summary, a facile and effective approach of U- and KOH-assisted thermal treatment was developed to introduce mesopores, cyano groups, and nitrogen vacancies into g-C_3N_4 simultaneously. The obtained thin-layered and porous *m*KCN exhibited a dramatic improvement in the photocatalytic activity for water reduction, with the optimized HER (1.557 mmol h^{-1} g^{-1}) under visible-light irradiation, which is about 48.5 times higher than that of pristine g-C_3N_4 synthesized by direct thermal polymerization. It is confirmed that the greatly enhanced photocatalytic activity can be mainly attributed to the synergistic effects of a larger specific surface area, improved light-harvesting capability, and enhanced photoexcited charge carrier separation efficiency. Besides, the formed in-plane mesopores are beneficial for the rapid mass-transfer process, resulting in largely improved reaction kinetics. Our findings demonstrate a synergistic modulation strategy of the structure and defects to design g-C_3N_4 with superb band structures and electronic structures for high-efficiency solar-energy capture and conversion. The strategy with tunable properties will open up further opportunities to modify 2D materials toward advanced applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.0c14262.

SEM images of MCN-B, MCN, M_xU_y, and *m*KCN; BJH pore-size distribution and the table of surface area pore parameters for MCN-B, MCN,10KMCN, M_1U_3, M_1U_4, and 10KCN; AFM images of the prepared samples; the XRD patterns and FTIR spectra of MCN-B, MCN, M_xU_y, and *m*KCN; the table of ^{13}C MAS NMR spectra peaks; the table of OEA data; the table of the XPS data including binding energies, atom ratios of the samples, and the atom contents of the samples determined from the ratios of peak areas; the high-resolution XPS spectra of C 1s and N 1s in MCN-B, MCN, 10KMCN-B, 10KMCN, M_1U_3-B, M_1U_3, 10KCN-B, and 10KCN; UV−vis diffuse reflectance spectra and plots of $(\alpha h\nu)^{1/2}$

versus E_g (eV) for MCN-B, MCN, M_xU_y, *m*KCN (*m* = 1, 5, 10, 20, 40, 100, and 200), and *m*KMCN (*m* = 10, 20, 40, 100, and 200), respectively; the VB-XPS spectra and band structures of samples; the calculated results of DFT for g-C_3N_4 with the cyano group and the N vacancy, respectively; the photoluminescence spectra and time-resolved fluorescence spectra of the samples; the table of the photoluminescence decay time of the samples derived from the time-resolved PL spectra; the electrochemical impedance spectra of the samples; the photocatalytic H_2 evolution rates and H_2 evolution curves of samples; the table of the photocatalytic activity of the 10KCN compared with the previous reported g-C_3N_4; the TEM micrograph, XRD patterns, and FTIR spectra of 10KCN before and after the photocatalytic reaction; and HPLC and LC/MS measurements of the aqueous solution (PDF)

AUTHOR INFORMATION

Corresponding Authors

Zhijie Wang − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China*; Email: wangzj@semi.ac.cn

Xiaobao Li − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China; School of Civil Engineering, Hefei University of Technology, Hefei 230009, China*; Email: xiaobaoli@hfut.edu.cn

Shengchun Qu − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China*; Email: qsc@semi.ac.cn

Authors

Yanbin Huang − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China; School of Mathematical Science and Engineering, Hebei University of Engineering, Handan 056038, China*

Jun Liu − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China*

Chao Zhao − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China*

Xiaohao Jia − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China*

Mengmeng Ma − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China*

Yuanyuan Qian − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China*

Cheng Yang − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China*

Kong Liu − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China*; orcid.org/0000-0002-1573-415X

Furui Tan − *Key Laboratory of Photovoltaic Materials, Department of Physics and Electronics, Henan University, Kaifeng 475004, Henan, China*

Zhanguo Wang − *Key Laboratory of Semiconductor Materials Science, Beijing Key Laboratory of Low Dimensional Semiconductor Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China; Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acsami.0c14262

Author Contributions

Y.H. and J.L. contributed equally to this work. The manuscript was written through the contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was mostly supported by the National Natural Science Foundation of China (Grant nos. 21975245, 51972300, and 61674141), the Key Research Program of Frontier Science, CAS (Grant no. QYZDB-SSW-SLH006), the National Key Research and Development Program of China (Grant nos. 2017YFA0206600 and 2018YFE0204000), and the Strategic Priority Research Program of Chinese Academy of Sciences (Grant no. XDB43000000). K.L. also acknowledges the support from the Youth Innovation Promotion Association, Chinese Academy of Sciences (no. 2020114), and the Natural Science Foundation of Hebei Province (F2019402063).

REFERENCES

(1) Chen, X.; Shen, S.; Guo, L.; Mao, S. S. Semiconductor-based Photocatalytic Hydrogen Generation. *Chem. Rev.* **2010**, *110*, 6503−6570.

(2) Wang, Z.; Li, C.; Domen, K. Recent developments in heterogeneous photocatalysts for solar-driven overall water splitting. *Chem. Soc. Rev.* **2019**, *48*, 2109−2125.

(3) Fujishima, A.; Honda, K. Electrochemical Photolysis of Water at a Semiconductor Electrode. *Nature* **1972**, *238*, 37−38.

(4) Kudo, A.; Miseki, Y. Heterogeneous photocatalyst materials for water splitting. *Chem. Soc. Rev.* **2009**, *38*, 253−278.

(5) Wang, Q.; Hisatomi, T.; Jia, Q.; Tokudome, H.; Zhong, M.; Wang, C.; Pan, Z.; Takata, T.; Nakabayashi, M.; Shibata, N.; Li, Y.; Sharp, I. D.; Kudo, A.; Yamada, T.; Domen, K. Scalable water splitting on particulate photocatalyst sheets with a solar-to-hydrogen energy conversion efficiency exceeding 1%. *Nat. Mater.* **2016**, *15*, 611−615.

(6) Kim, J. H.; Hansora, D.; Sharma, P.; Jang, J. W.; Lee, J. S. Toward practical solar hydrogen production-an artificial photosynthetic leaf-to-farm challenge. *Chem. Soc. Rev.* **2019**, *48*, 1908−1971.

(7) Huang, Y.; Liu, J.; Deng, Y.; Qian, Y.; Jia, X.; Ma, M.; Yang, C.; Liu, K.; Wang, Z.; Qu, S.; Wang, Z. The application of perovskite materials in solar water splitting. *J. Semicond.* **2020**, *41*, No. 011701.

(8) Nasori, N.; Dai, T.; Jia, X.; Rubiyanto, A.; Cao, D.; Qu, S.; Wang, Z.; Wang, Z.; Lei, Y. Realizing super-long Cu_2O nanowires arrays for high-efficient water splitting applications with a convenient approach. *J. Semicond.* **2019**, *40*, No. 052701.

(9) Huang, Y.; Liu, J.; Cao, D.; Zheming, L.; Ren, K.; Liu, K.; Tang, A.; Wang, Z.; Li, L.; Qu, S.; Wang, Z. Separation of hot electrons and holes in Au/$LaFeO_3$ to boost the photocatalytic activities both for water reduction and oxidation. *Int. J. Hydrogen Energy* **2019**, *44*, 13242−13252.

(10) Wang, X.; Maeda, K.; Thomas, A.; Takanabe, K.; Xin, G.; Carlsson, J. M.; Domen, K.; Antonietti, M. A metal-free polymeric photocatalyst for hydrogen production from water under visible light. *Nat. Mater.* **2009**, *8*, 76−80.

(11) Cao, S.; Low, J.; Yu, J.; Jaroniec, M. Polymeric Photocatalysts Based on Graphitic Carbon Nitride. *Adv. Mater.* **2015**, *27*, 2150−2176.

(12) Lau, V. W.-h.; Mesch, M. B.; Duppel, V.; Blum, V.; Senker, J.; Lotsch, B. V. Low-Molecular-Weight Carbon Nitrides for Solar Hydrogen Evolution. *J. Am. Chem. Soc.* **2015**, *137*, 1064−1072.

(13) Liao, G.; Gong, Y.; Zhang, L.; Gao, H.; Yang, G.-J.; Fang, B. Semiconductor polymeric graphitic carbon nitride photocatalysts: the "holy grail" for the photocatalytic hydrogen evolution reaction under visible light. *Energy Environ. Sci.* **2019**, *12*, 2080−2147.

(14) Yang, P.; Zhuzhang, H.; Wang, R.; Lin, W.; Wang, X. Carbon Vacancies in a Melon Polymeric Matrix Promote Photocatalytic Carbon Dioxide Conversion. *Angew. Chem.* **2019**, *131*, 1146−1149.

(15) Wang, H.; Jiang, S.; Chen, S.; Li, D.; Zhang, X.; Shao, W.; Sun, X.; Xie, J.; Zhao, Z.; Zhang, Q.; Tian, Y.; Xie, Y. Enhanced Singlet Oxygen Generation in Oxidized Graphitic Carbon Nitride for Organic Synthesis. *Adv. Mater.* **2016**, *28*, 6940−6945.

(16) Li, Y.; Ouyang, S.; Xu, H.; Wang, X.; Bi, Y.; Zhang, Y.; Ye, J. Constructing Solid−Gas-Interfacial Fenton Reaction over Alkalinized-C_3N_4 Photocatalyst To Achieve Apparent Quantum Yield of 49% at 420 nm. *J. Am. Chem. Soc.* **2016**, *138*, 13289−13297.

(17) Ong, W.-J.; Tan, L.-L.; Ng, Y. H.; Yong, S.-T.; Chai, S.-P. Graphitic Carbon Nitride (g-C_3N_4)-Based Photocatalysts for Artificial Photosynthesis and Environmental Remediation: Are We a Step Closer To Achieving Sustainability? *Chem. Rev.* **2016**, *116*, 7159−7329.

(18) Zhao, Y.; Zhao, F.; Wang, X.; Xu, C.; Zhang, Z.; Shi, G.; Qu, L. Graphitic Carbon Nitride Nanoribbons: Graphene-Assisted Formation and Synergic Function for Highly Efficient Hydrogen Evolution. *Angew. Chem., Int. Ed.* **2014**, *53*, 13934−13939.

(19) Yu, H.; Shi, R.; Zhao, Y.; Bian, T.; Zhao, Y.; Zhou, C.; Waterhouse, G. I. N.; Wu, L.-Z.; Tung, C.-H.; Zhang, T. Alkali-Assisted Synthesis of Nitrogen Deficient Graphitic Carbon Nitride with Tunable Band Structures for Efficient Visible-Light-Driven Hydrogen Evolution. *Adv. Mater.* **2017**, *29*, No. 1605148.

(20) Wang, Y.; Du, P.; Pan, H.; Fu, L.; Zhang, Y.; Chen, J.; Du, Y.; Tang, N.; Liu, G. Increasing Solar Absorption of Atomically Thin 2D Carbon Nitride Sheets for Enhanced Visible-Light Photocatalysis. *Adv. Mater.* **2019**, *31*, No. 1807540.

(21) Hong, Y.; Wang, L.; Liu, E.; Chen, J.; Wang, Z.; Zhang, S.; Lin, X.; Duan, X.; Shi, J. A curly architectured graphitic carbon nitride (g-C_3N_4) towards efficient visible-light photocatalytic H_2 evolution. *Inorg. Chem. Front.* **2020**, *7*, 347−355.

(22) Wang, L.; Hong, Y.; Liu, E.; Wang, Z.; Chen, J.; Yang, S.; Wang, J.; Lin, X.; Shi, J. Rapid polymerization synthesizing high-crystalline g-C_3N_4 towards boosting solar photocatalytic H_2 generation. *Int. J. Hydrogen Energy* **2020**, *45*, 6425−6436.

(23) Jiang, L.; Yuan, X.; Pan, Y.; Liang, J.; Zeng, G.; Wu, Z.; Wang, H. Doping of graphitic carbon nitride for photocatalysis: A reveiw. *Appl. Catal., B* **2017**, *217*, 388−406.

(24) Meng, N.; Ren, J.; Liu, Y.; Huang, Y.; Petit, T.; Zhang, B. Engineering oxygen-containing and amino groups into two-dimensional atomically-thin porous polymeric carbon nitrogen for enhanced photocatalytic hydrogen production. *Energy Environ. Sci.* **2018**, *11*, 566−571.

(25) Zhao, D.; Dong, C.-L.; Wang, B.; Chen, C.; Huang, Y.-C.; Diao, Z.; Li, S.; Guo, L.; Shen, S. Synergy of Dopants and Defects in Graphitic Carbon Nitride with Exceptionally Modulated Band Structures for Efficient Photocatalytic Oxygen Evolution. *Adv. Mater.* **2019**, *31*, No. 1903545.

(26) Guo, F.; Li, M.; Ren, H.; Huang, X.; Shu, K.; Shi, W.; Lu, C. Facile bottom-up preparation of Cl-doped porous g-C_3N_4 nanosheets for enhanced photocatalytic degradation of tetracycline under visible light. *Sep. Purif. Technol.* **2019**, *228*, No. 115770.

(27) Guo, F.; Wang, L.; Sun, H.; Li, M.; Shi, W. High-efficiency photocatalytic water splitting by a N-doped porous g-C_3N_4 nanosheet polymer photocatalyst derived from urea and N,N-dimethylformamide. *Inorg. Chem. Front.* **2020**, *7*, 1770−1779.

(28) Shi, W.; Li, M.; Huang, X.; Ren, H.; Yan, C.; Guo, F. Facile synthesis of 2D/2D $Co_3(PO_4)_2$/ g-C_3N_4 heterojunction for highly photocatalytic overall water splitting under visible light. *Chem. Eng. J.* **2019**, *382*, No. 122960.

(29) Yang, S.; Gong, Y.; Zhang, J.; Zhan, L.; Ma, L.; Fang, Z.; Vajtai, R.; Wang, X.; Ajayan, P. M. Exfoliated Graphitic Carbon Nitride Nanosheets as Efficient Catalysts for Hydrogen Evolution Under Visible Light. *Adv. Mater.* **2013**, *25*, 2452−2456.

(30) Li, Y.; Ouyang, S.; Xu, H.; Hou, W.; Zhao, M.; Chen, H.; Ye, J. Targeted Exfoliation and Reassembly of Polymeric Carbon Nitride for Efficient Photocatalysis. *Adv. Funct. Mater.* **2019**, *29*, No. 1901024.

(31) Zhang, L.; Ding, N.; Lou, L.; Iwasaki, K.; Wu, H.; Luo, Y.; Li, D.; Nakata, K.; Fujishima, A.; Meng, Q. Localized Surface Plasmon Resonance Enhanced Photocatalytic Hydrogen Evolution via Pt@Au NRs/C_3N_4 Nanotubes under Visible-Light Irradiation. *Adv. Funct. Mater.* **2019**, *29*, No. 1806774.

(32) Marchal, C.; Cottineau, T.; Méndez-Medrano, M. G.; Colbeau-Justin, C.; Caps, V.; Keller, V. Au/TiO_2−gC_3N_4 Nanocomposites for

https://dx.doi.org/10.1021/acsami.0c14262
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

Enhanced Photocatalytic H_2 Production from Water under Visible Light Irradiation with Very Low Quantities of Sacrificial Agents. *Adv. Energy Mater.* **2018**, *8*, No. 1702142.
(33) Ba, G.; Liang, Z.; Li, H.; Du, N.; Liu, J.; Hou, W. Simultaneous formation of mesopores and homojunctions in graphite carbon nitride with enhanced optical absorption, charge separation and photocatalytic hydrogen evolution. *Appl. Catal., B* **2019**, *253*, 359−368.
(34) Shi, W.; Liu, C.; Li, M.; Lin, X.; Guo, F.; Shi, J. Fabrication of ternary $Ag_3PO_4/Co_3(PO_4)_2$/g-C_3N_4 heterostructure with following Type II and Z-Scheme dual pathways for enhanced visible-light photocatalytic activity. *J. Hazard. Mater.* **2020**, *389*, No. 121907.
(35) Deng, Y.; Liu, J.; Huang, Y.; Ma, M.; Liu, K.; Dou, X.; Wang, Z.; Qu, S.; Wang, Z. Engineering the Photocatalytic Behaviors of g/C_3N_4-Based Metal-Free Materials for Degradation of a Representative Antibiotic. *Adv. Funct. Mater.* **2020**, *30*, No. 2002353.
(36) Guo, F.; Huang, X.; Chen, Z.; Sun, H.; Chen, L. Prominent co-catalytic effect of CoP nanoparticles anchored on high-crystalline g-C_3N_4 nanosheets for enhanced visible-light photocatalytic degradation of tetracycline in wastewater. *Chem. Eng. J.* **2020**, *395*, No. 125118.
(37) Han, Q.; Cheng, Z.; Wang, B.; Zhang, H.; Qu, L. Significant Enhancement of Visible-Light-Driven Hydrogen Evolution by Structure Regulation of Carbon Nitrides. *ACS Nano* **2018**, *12*, 5221−5227.
(38) Hong, Y.; Liu, E.; Shi, J.; Lin, X.; Sheng, L.; Zhang, M.; Wang, L.; Chen, J. A direct one-step synthesis of ultrathin g-C_3N_4 nanosheets from thiourea for boosting solar photocatalytic H_2 evolution. *Int. J. Hydrogen Energy* **2019**, *44*, 7194−7204.
(39) Zhang, D.; Guo, Y.; Zhao, Z. Porous defect-modified graphitic carbon nitride via a facile one-step approach with significantly enhanced photocatalytic hydrogen evolution under visible light irradiation. *Appl. Catal., B* **2018**, *226*, 1−9.
(40) Wang, Y.; Liu, X.; Zheng, C.; Li, Y.; Jia, S.; Li, Z.; Zhao, Y. Tailoring TiO_2 Nanotube-Interlaced Graphite Carbon Nitride Nanosheets for Improving Visible-Light-Driven Photocatalytic Performance. *Adv. Sci.* **2018**, *5*, No. 1700844.
(41) Ran, J.; Ma, T. Y.; Gao, G.; Du, X.-W.; Qiao, S. Z. Porous P-doped graphitic carbon nitride nanosheets for synergistically enhanced visible-light photocatalytic H_2 production. *Energy Environ. Sci.* **2015**, *8*, 3708−3717.
(42) Kresse, G.; Hafner, J. Ab initio molecular dynamics for liquid metals. *Phys. Rev. B: Condens. Matter* **1993**, *47*, 558−561.
(43) Kresse, G.; Furthmuller, J. Efficient Iterative Schemes for Ab Initio Total-Energy Calculations Using a Plane-Wave Basis Set. *Phys. Rev. B* **1996**, *54*, 11169−11186.
(44) Blöchl, P. E. Projector augmented-wave method. *Phys. Rev. B: Condens. Matter* **1994**, *50*, 17953−17979.
(45) Monkhorst, H. J.; Pack, J. D. Special points for Brillouin-zone integrations. *Phys. Rev. B* **1976**, *13*, 5188−5192.
(46) Ruan, D.; Kim, S.; Fujitsuka, M.; Majima, T. Defects rich g-C_3N_4 with mesoporous structure for efficient photocatalytic H_2 production under visible light irradiation. *Appl. Catal., B* **2018**, *238*, 638−646.
(47) Hu, P.; Chen, C.; Zeng, R.; Xiang, J.; Huang, Y.; Hou, D.; Li, Q.; Huang, Y. Facile synthesis of bimodal porous graphitic carbon nitride nanosheets as efficient photocatalysts for hydrogen evolution. *Nano Energy* **2018**, *50*, 376−382.
(48) Qie, L.; Chen, W.; Xu, H.; Xiong, X.; Jiang, Y.; Zou, F.; Hu, X.; Xin, Y.; Zhang, Z.; Huang, Y. Synthesis of functionalized 3D hierarchical porous carbon for high-performance supercapacitors. *Energy Environ. Sci.* **2013**, *6*, 2497−2504.
(49) He, F.; Chen, G.; Yu, Y.; Zhou, Y.; Zheng, Y.; Hao, S. The sulfur-bubble template-mediated synthesis of uniform porous g-C_3N_4 with superior photocatalytic performance. *Chem. Commun.* **2015**, *51*, 425−427.
(50) Tian, N.; Xiao, K.; Zhang, Y.; Lu, X.; Ye, L.; Gao, P.; Ma, T.; Huang, H. Reactive sites rich porous tubular yolk-shell g-C_3N_4 via precursor recrystallization mediated microstructure engineering for photoreduction. *Appl. Catal., B* **2019**, *253*, 196−205.
(51) Gao, H.; Yan, S.; Wang, J.; Huang, Y. A.; Wang, P.; Li, Z.; Zou, Z. Towards efficient solar hydrogen production by intercalated carbon nitride photocatalyst. *Phys. Chem. Chem. Phys.* **2013**, *15*, 18077−18084.
(52) Jürgens, B.; Irran, E.; Senker, J.; Kroll, P.; Müller, H.; Schnick, W. Melem (2,5,8-Triamino-tri-s-triazine), an Important Intermediate during Condensation of Melamine Rings to Graphitic Carbon Nitride: Synthesis, Structure Determination by X-ray Powder Diffractometry, Solid-State NMR, and Theoretical Studies. *J. Am. Chem. Soc.* **2003**, *125*, 10288−10300.
(53) Sano, T.; Tsutsui, S.; Koike, K.; Hirakawa, T.; Teramoto, Y.; Negishi, N.; Takeuchi, K. Activation of graphitic carbon nitride (g-C_3N_4) by alkaline hydrothermal treatment for photocatalytic NO oxidation in gas phase. *J. Mater. Chem. A* **2013**, *1*, 6489−6496.
(54) Zhou, P.; Lv, F.; Li, N.; Zhang, Y.; Mu, Z.; Tang, Y.; Lai, J.; Chao, Y.; Luo, M.; Lin, F.; Zhou, J.; Su, D.; Guo, S. Strengthening reactive metal-support interaction to stabilize high-density Pt single atoms on electron-deficient g-C_3N_4 for boosting photocatalytic H_2 production. *Nano Energy* **2019**, *56*, 127−137.
(55) Chen, Z.; Vorobyeva, E.; Mitchell, S.; Fako, E.; Ortuño, M. A.; López, N.; Collins, S. M.; Midgley, P. A.; Richard, S.; Vilé, G.; Pérez-Ramírez, J. A heterogeneous single-atom palladium catalyst surpassing homogeneous systems for Suzuki coupling. *Nat. Nanotechnol.* **2018**, *13*, 702−707.
(56) Zhang, X.; Xie, X.; Wang, H.; Zhang, J.; Pan, B.; Xie, Y. Enhanced Photoresponsive Ultrathin Graphitic-Phase C_3N_4 Nanosheets for Bioimaging. *J. Am. Chem. Soc.* **2013**, *135*, 18−21.
(57) Chen, X.; Liu, L.; Yu, P. Y.; Mao, S. S. Increasing Solar Absorption for Photocatalysis with Black Hydrogenated Titanium Dioxide Nanocrystals. *Science* **2011**, *331*, 746−750.
(58) Tu, W.; Xu, Y.; Wang, J.; Zhang, B.; Zhou, T.; Yin, S.; Wu, S.; Li, C.; Huang, Y.; Zhou, Y.; Zou, Z.; Robertson, J.; Kraft, M.; Xu, R. Investigating the Role of Tunable Nitrogen Vacancies in Graphitic Carbon Nitride Nanosheets for Efficient Visible-Light-Driven H_2 Evolution and CO_2 Reduction. *ACS Sustainable Chem. Eng.* **2017**, *5*, 7260−7268.
(59) Wang, J.; Yang, Z.; Yao, W.; Gao, X.; Tao, D. Defects modified in the exfoliation of g-C_3N_4 nanosheets via a self-assembly process for improved hydrogen evolution performance. *Appl. Catal., B* **2018**, *238*, 629−637.
(60) Zhu, Y.; Marianov, A.; Xu, H.; Lang, C.; Jiang, Y. Bimetallic Ag-Cu Supported on Graphitic Carbon Nitride Nanotubes for Improved Visible-Light Photocatalytic Hydrogen Production. *ACS Appl. Mater. Interfaces* **2018**, *10*, 9468−9477.
(61) Hollmann, D.; Karnahl, M.; Tschierlei, S.; Kailasam, K.; Schneider, M.; Radnik, J.; Grabow, K.; Bentrup, U.; Junge, H.; Beller, M.; Lochbrunner, S.; Thomas, A.; Brückner, A. Structure-Activity Relationships in Bulk Polymeric and Sol-Gel-Derived Carbon Nitrides during Photocatalytic Hydrogen Production. *Chem. Mater.* **2014**, *26*, 1727−1733.
(62) Zhang, G.; Zhang, M.; Ye, X.; Qiu, X.; Lin, S.; Wang, X. Iodine Modified Carbon Nitride Semiconductors as Visible Light Photocatalysts for Hydrogen Evolution. *Adv. Mater.* **2014**, *26*, 805−809.